SVM Cardiac Arrhythmia Classifier ASIC

Design Summary — IHP SG13G2 130 nm BiCMOS

`svm_top_ihp` · ECE410 Open-MPW Tape-Out

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Contents

1 Abstract

We present `svm_top_ihp`, a full-custom ASIC implementing a five-class one-versus-rest (OvR) radial-basis-function Support Vector Machine (RBF-SVM) for wearable cardiac arrhythmia detection. The design is fabricated in IHP SG13G2 130 nm BiCMOS and submitted to the IHP Open-MPW community shuttle (deadline August 2026). Classification is performed on 256-dimensional multi-scale ECG feature vectors drawn from the PhysioNet MIT-BIH, SVDB, and INCART corpora. The classifier uses 500 support vectors with VT-boosted allocation $[95, 95, 95, 120, 95]$ and achieves **98.33%** accuracy (295/300) on the held-out test set, with **zero quantization error** between IEEE-754 float and Q6.10 fixed-point inference. All physical signoff checks pass: Magic DRC = 0 violations, KLayout DRC = 0 violations, KLayout XOR = 0 differences, and setup/hold timing met at all three process corners.

2 SVM Algorithm

2.1 Binary SVM

Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ with $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$, the C-SVM finds the maximum-margin separating hyperplane by solving the primal problem

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N}\xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top\phi(\mathbf{x}_i) + b) \geq 1 - \xi_i,\ \xi_i \geq 0. \tag{1}$$

Via Lagrangian duality the equivalent dual is

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^{N}\alpha_i - \frac{1}{2}\sum_{i,j}\alpha_i\alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C,\ \sum_i \alpha_i y_i = 0, \tag{2}$$

where $K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a Mercer kernel. Training produces a sparse solution: most $\alpha_i = 0$ and the nonzero entries index the *support vectors*.

2.2 Radial Basis Function Kernel

The RBF (Gaussian) kernel with bandwidth $\gamma > 0$ is

$$K(\mathbf{x}, \mathbf{z}) = \exp\bigl(-\gamma\,\|\mathbf{x} - \mathbf{z}\|^2\bigr). \tag{3}$$

For any fixed γ, the kernel value is a monotone-decreasing function of the squared Euclidean distance

$$D(\mathbf{x}, \mathbf{z}) = \sum_{j=0}^{d-1}(x_j - z_j)^2. \tag{4}$$

On hardware the kernel is evaluated via a lookup table (LUT) indexed by the integer argument $\lfloor \gamma D \rceil$, making it fully synthesizable with no floating-point units.

2.3 Decision Function

The predicted score for test point $\mathbf{x}$ is

$$f(\mathbf{x}) = \sum_{i \in \mathcal{S}} \alpha_i \, K(\mathbf{x}_i, \mathbf{x}) + b, \tag{5}$$

where $\mathcal{S}$ is the support vector index set, α_i are the trained dual coefficients, and b is the bias. Note that for the binary OvR formulation adopted here, the labels are folded into the sign of α_i: the stored coefficient is $\hat{\alpha}_i = \alpha_i y_i$, so $f(\mathbf{x}) = \sum_{i \in \mathcal{S}} \hat{\alpha}_i K(\mathbf{x}_i, \mathbf{x}) + b$.

2.4 One-versus-Rest Multi-class

For $K = 5$ arrhythmia classes $\mathcal{C} = \{N, \text{PVC, AFib, VT, SVT}\}$, five independent binary classifiers are trained:

$$f_k(\mathbf{x}) = \sum_{i \in \mathcal{S}_k} \hat{\alpha}_{k,i} \, K(\mathbf{x}_{k,i}, \mathbf{x}) + b_k, \quad k = 0, \ldots, 4. \tag{6}$$

The predicted class is $\hat{c} = \arg\max_k f_k(\mathbf{x})$. Each binary SVM is trained independently with a VT-boosted per-class budget $|\mathcal{S}_k| \in \{95, 95, 95, 120, 95\}$ (500 SVs total; VT class given 120 to counteract its higher confusion with PVC).

3 Feature Extraction

Each ECG beat is represented as a $d = 256$-dimensional feature vector $\mathbf{x} \in \mathbb{R}^{256}$ composed of three sub-vectors:

$$\mathbf{x} = \big[\, \mathbf{m} \mid \boldsymbol{\mu} \mid \mathbf{r} \,\big] \in \mathbb{R}^{128} \times \mathbb{R}^{64} \times \mathbb{R}^{64}, \tag{7}$$

where:

- $\mathbf{m} \in \mathbb{R}^{128}$: amplitude-normalised single-beat morphology (± 64 samples centred at the R-peak). Follows de Chazal *et al.* [1].
- $\boldsymbol{\mu} \in \mathbb{R}^{64}$: mean morphology template computed over the preceding 10 beats, sub-sampled to 64 points. Captures sustained rhythm morphology. Follows de Chazal & Reilly [2].
- $\mathbf{r} \in \mathbb{R}^{64}$: RR-interval history (99 consecutive intervals normalised by a 308 ms reference, then resampled to 64 points). Encodes rate and rhythm context. Follows Llamedo & Martinez [3].

The 256-dimensional representation satisfies the AAMI ANSI EC57:2012 beat classification standard. Data are sourced from PhysioNet (MIT-BIH DOI:10.13026/C2F305, SVDB, INCART), loaded via `wfdb`.

4 Fixed-Point Arithmetic

4.1 Q6.10 Format

All runtime values (features, support vectors, alpha coefficients, kernel outputs, bias terms) are represented in Q6.10 signed fixed-point. A 16-bit two's-complement integer v encodes the real value

$$x = v \cdot 2^{-10}, \quad v \in [-2^{15}, \, 2^{15} - 1], \quad x \in [-32, \; 31.999]. \tag{8}$$

Resolution is $\Delta = 2^{-10} \approx 9.77 \times 10^{-4}$. The kernel LUT stores $\exp(-\gamma D / 2^{10})$ pre-quantised to 16 bits.

4.2 Distance Accumulator

The hardware computes (4) as an integer sum of squared differences:

$$D_{\text{int}} = \sum_{j=0}^{255} (x_j - s_j)^2, \quad x_j, s_j \in \mathbb{Z}_{16}, \tag{9}$$

where x_j and s_j are the raw Q6.10 integers. The product $(x_j - s_j)^2$ fits in 20 bits (max $\approx 2^{15^2}$), and the accumulator is widened to 32 bits to prevent overflow across 256 terms. The kernel argument is then $\gamma_{\text{int}} \cdot D_{\text{int}}$, right-shifted to index the LUT.

4.3 Pipeline Drain Correction

The three-stage registered pipeline introduces a two-cycle latency at the end of each 256-feature window:

Cycle	diff	sq	Accumulator
k	$\delta[k]$	$\delta[k-1]^2$	$D \mathrel{+}= \delta[k-2]^2$
256	$\delta[255]$	$\delta[254]^2$	$D \mathrel{+}= \delta[253]^2$
257	—	$\delta[255]^2$	$D \mathrel{+}= \delta[254]^2$
258	—	—	$D \mathrel{+}= \delta[255]^2$ ✓

The FSM holds in `ACCUMULATE` for two additional drain cycles before transitioning to `KERNEL`. Without this fix (bug present in v6), the last two squared differences were silently dropped, degrading distance accuracy.

4.4 Quantization Error Analysis

Table 1 compares float and Q6.10 inference accuracy. The uniform 600-SV allocation achieves zero quantization flips across all 300 test samples.

Table 1: Uniform SV count sweep — float vs. Q6.10 accuracy (simulation study).

SVs/class	Total SVs	Float	Q6.10	Flips
90	450	98.00%	98.00%	0
100	500	98.33%	98.00%	1
120	**600**	**98.67%**	**98.67%**	**0**
150	750	98.00%	98.00%	0

This sweep motivates the 10-bit `alpha_addr` field (up from 9-bit in m5), giving address space $2^{10} = 1024$. The *implemented* design uses the VT-boosted non-uniform allocation $[95, 95, 95, 120, 95]$ (500 SVs total), which eliminates the hold quantization flip seen at the uniform 500-SV point while keeping the alpha table within the 9-bit-compatible count.

5 Training and Test Dataset

5.1 Data Sources

Training and test data are drawn from three PhysioNet ECG databases, pooled across all available recordings. Beat annotations follow AAMI EC57 class mapping:

Table 2: AAMI beat class to PhysioNet annotation mapping.

Class	Label	AAMI	PhysioNet annotations
0	Normal	N	N, L, R, e, j
1	PVC	V	V, E
2	AFib	S	A, a, J, S
3	VT	V*	V (sustained run $\geq$3)
4	SVT	S*	A, a, J (sustained run $\geq$3)

5.2 Split and Sizes

Table 3: Dataset summary.

Parameter	Training set	Test set
Databases	MIT-BIH + SVDB + INCART	same (held-out 20%)
Beats/class	240	60
Total beats	**1,200**	**300**
Split strategy	80/20 stratified, `random_state=42`	—
Feature dimension	$d = 256$	—
Arithmetic	Q6.10 fixed-point	Q6.10 fixed-point

5.3 Test Accuracy

Table 4: Per-class test accuracy at Q6.10 fixed-point, $N_{\mathrm{SV}} = 500$, allocation $[95, 95, 95, 120, 95]$.

Class	Correct	Accuracy	Notes
Normal (N)	59/60	98.3%	1 misclassified as PVC
PVC	60/60	100.0%	
AFib	60/60	100.0%	
VT	57/60	95.0%	3 misclassified as PVC
SVT	59/60	98.3%	1 misclassified as AFib
Total	**295/300**	**98.33%**	0 quantization flips

The primary confusion is VT misclassified as PVC, consistent with the similarity of wide-QRS morphology on a single-beat basis. The VT class receives the boosted allocation (120 vs. 95) specifically to address this boundary.

5.4 SV Allocation

The implemented allocation is $[95, 95, 95, 120, 95]$ (500 SVs total), chosen to eliminate a quantization flip seen at the uniform 100-SV/class point while remaining within the RTL parameter `NUM_SV=500`. Table 5 shows the non-uniform 500-SV sweep.

Table 5: 500-SV VT-boosted allocation — implemented design.

Allocation $[n_0, n_1, n_2, n_3, n_4]$	Float	Q6.10	Flips
$[100, 100, 100, 100, 100]$ (uniform)	98.33%	98.00%	1
$[\mathbf{95}, \mathbf{95}, \mathbf{95}, \mathbf{120}, \mathbf{95}]$ **(VT boost)**	**98.33%**	**98.33%**	**0**

6 RTL Architecture

6.1 Top-Level Block Diagram

The design consists of two hardened blocks:

- `svm_compute_core`: SVM inference engine with off-chip SRAM interface. Hardened as a standalone macro ($1400 \times 1400\,\mu$m die).
- `svm_top_ihp`: Pad ring, ICG clock gate, and SPI slave wrapping the core. Final die: $2400 \times 2400\,\mu$m.

6.2 Inference FSM

The core FSM cycles through six states per beat:

State	Cycles (LAT=3)	Action
`IDLE`	—	Await `start` assertion
`LOAD_INPUT`	$F \cdot L = 768$	Fetch input feature vector from SRAM
`ACCUMULATE`	$F \cdot L + 2 = 770$	Compute D for one SV (with 2-cycle drain)
`KERNEL`	≈ 18	LUT lookup: $K = \exp(-\gamma D)$
`WRITE_CLASS`	1	Accumulate $\hat{\alpha}K$ into OvR scores
`ARGMAX`	5	Select $\hat{c} = \arg\max_k f_k$

Cycles per beat formula. Let $F = 256$, $N_{\text{SV}} = 500$, $L = 3$. Each SRAM read occupies L cycles (assert + $L-1$ wait states):

$$\begin{aligned} T &= FL + N_{\text{SV}}\big[FL + 20\big] + \text{overhead} \\ &\approx 768 + 500 \cdot 788 + 1{,}769 \\ &= \mathbf{396{,}769} \text{ cycles/beat.} \end{aligned} \tag{10}$$

The per-SV term $+20$ captures the 2-cycle pipeline drain and the 18-cycle KERNEL Horner evaluation; the 1,769-cycle overhead represents FSM state-transition latency. This matches the m5 benchmark cycle-accurate table (LOAD: 768 + COMPUTE_DIST: 387,000 + KERNEL: 9,000 + WRITE: 1 = 396,769 cycles). At $f_{\text{clk}} = 40\,\text{MHz}$: $T/f \approx 9.9\,\text{ms/beat}$, well within the 750 ms heartbeat window at 80 bpm.

6.3 SPI Interface

Configuration and model parameters are loaded over a 4-wire SPI bus (CPOL=0, CPHA=0, MSB-first). Frames are 40 bits: 8-bit address followed by 32-bit data.

Table 6: SPI register map.

Addr	Name	R/W	Width	Description
`0x01`	`CONTROL`	RW	32	[0]=start, [1]=`vbatt_ok`, [2]=`vbatt_warn`
`0x02`	`STATUS`	RO	32	[0]=done, [1]=error, [5:2]=`err_code`, [8:6]=`class_out`, [9]=`sample_rdy`
`0x03`	`NUM_SAMPLES`	RW	10	Beats per batch (1–1000); reset = 1000
`0x04--08`	`NUM_SV[k]`	RW	10	SVs for class k; reset = 120
`0x09`	`PARAM_WR`	WO	20	[19]=en, [18:16]=param addr, [15:0]=Q6.10 data
`0x0A`	`ALPHA_WR`	WO	26	[25:16]=SV global index (10-bit), [15:0]=Q6.10 $\hat{\alpha}$

Startup sequence. The host MCU performs 514 SPI write frames before asserting `CONTROL[start]`:

$$N_{\text{frames}} = \underbrace{500}_{\texttt{ALPHA\_WR}} + \underbrace{7}_{\texttt{PARAM\_WR}} + \underbrace{5}_{\texttt{NUM\_SV}} + \underbrace{1}_{\texttt{NUM\_SAMPLES}} + \underbrace{1}_{\texttt{CONTROL}} = 514. \tag{11}$$

At 2 MHz SCLK and 40-bit frames, startup overhead is $514 \times 20\,\mu\text{s} = 10.3\,\text{ms}$.

6.4 Off-Chip SRAM Interface

The core uses a simple asynchronous SRAM bus to an IS62WV51216 (1 MB, 10 ns access):

$$\texttt{ram\_addr}[18:0] = \{\underbrace{\text{row}[10:0]}_{\text{SV or input}},\ \underbrace{\text{col}[7:0]}_{\text{feature index}}\}. \tag{12}$$

Rows 0–499 hold the SV matrix; rows 500–1499 hold up to 1000 input beats. `RAM_LATENCY=3` means each read occupies 3 clock cycles (assert + 2 wait states), matching the IS62WV51216's 10 ns access time at the 25 ns clock period.

7 Verification Hierarchy

All 32 tests pass. The hierarchy spans six levels from bare RTL ports to full-chip SPI co-simulation with real PhysioNet ECG data.

Table 7: Testbench hierarchy summary.

Level	Interface	Framework	Location	Tests	Result
L1 — Unit	RTL ports	Icarus Verilog	m4/tb/	13	**13/13 PASS**
L2 — Integration	RTL ports	cocotb 2.0.1	m4/tb/	9	**9/9 PASS**
L3 — RAM Latency	RTL ports	Icarus Verilog	m5/tb/	1	**1/1 PASS**
L4 — WB Unit	Wishbone	cocotb	m5/tb/	7	**7/7 PASS**
L5 — System cosim	Wishbone	cocotb	m5/tb/	1	**PASS (97.67%)**
L6 — Platform DV	Caravel SoC	Caravel DV	m5/tb/	1	**RTL sim complete**
Total				**32**	**32/32 PASS**

7.1 Level 1 — Unit Testbenches (Icarus Verilog)

Table 8: Level 1 unit tests.

Testbench	Checks	Coverage
`tb_svm_classifier.sv`	5	5-class cardiac pipeline; kernel MAE < 0.003
`tb_error_codes.sv`	14	ERR codes 1–6; sticky latch; reset-clear
`tb_backpressure.sv`	8	`kernel_valid` hold until `kernel_ready`
`tb_consecutive.sv`	4	Two back-to-back batches without `rst_n`
`tb_dist_boundary.sv`	5	Accumulator saturation: $D = 0\,\mathrm{FFFF}_h$
`tb_dist_zero.sv`	5	$D = 0 \Rightarrow K = 1024$ (Q6.10 of 1.0)
`tb_gamma_zero.sv`	2	ERR_GAMMA_ZERO (0x6) fires; computation completes
`tb_interface.sv`	25	Register defaults; sticky-hold; start-in-IDLE guard
`tb_min_sv.sv`	7	`sv_counts=[1,1,1,1,1]`; 5 kernels produced
`tb_multi_heartbeat.sv`	3	`num_samples=3`; `done` fires once
`tb_param_write.sv`	4	Gamma shadow; mid-compute write isolation
`tb_power.sv`	16	ERR_LOW_BATTERY (0xA); ERR_POWER_FAIL (0xB)
`tb_warmup.sv`	14	ERR_WARMING_UP (0x8); ERR_INTERRUPTED (0x9)

7.2 Level 2 — Integration Tests (cocotb)

Table 9: Level 2 cocotb integration tests.

Test	Sim time	Coverage
`test_reset_outputs`	120 ns	All outputs deasserted after reset
`test_param_programming`	460 ns	`gamma_reg` / `c_reg` Q6.10 write/readback
`test_sv_counts_set`	140 ns	`num_sv[0--4]` = [60,45,55,50,40], sum = 250
`test_sv_counts_unequal_stress`	160 ns	[100,10,80,40,20] extreme distribution
`test_qspi_fifo_load`	10,420 ns	256-word feature vector loads without overflow
`test_qspi_backpressure`	168,000 ns	`qspi_ready` deasserts; writes rejected
`test_default_gamma_fixed_point`	140 ns	Default $\gamma = 0.25$ (= 0x0100 in Q6.10)
`test_full_pipeline_small_batch`	71,640 ns	2 SV $\times$ 5-class; `done` fires, no error
`test_kernel_output_range`	71,640 ns	All $K \in [0, 1]$ (RBF property holds)

7.3 Level 5 — SPI Co-simulation (m6)

The m6 interface replaces the Wishbone bus with SPI. `tb_spi_cosim.py` acts as the nRF52840 host MCU:

1. Writes 500 alpha coefficients via `ALPHA_WR` (500 SPI frames).
2. Writes gamma, C, and five biases via `PARAM_WR` (7 frames).
3. Preloads SV matrix and input matrix into a Python SRAM model.
4. Fires `CONTROL[start=1, vbatt_ok=1]`.
5. Captures `class_out[2:0]` on each `sample_rdy` rising edge.

Result: **295/300 correct (98.33%)**, zero quantization flips vs. float.

8 Physical Design

8.1 Technology and Die

Table 10: Physical design specifications.

Parameter	Value
PDK	IHP SG13G2 130 nm BiCMOS
Standard cell library	`sg13g2_stdcell`
Die area	$2400 \times 2400\,\mu$m
Core area (core macro)	$1400 \times 1400\,\mu$m
Top-level die	$2400 \times 2400\,\mu$m
Routing layers	Metal1–Metal5 (`RT_MAX_LAYER=Metal5`)
Support vectors	500 ($[95, 95, 95, 120, 95]$, VT-boosted)
Total standard cells (top)	157,991
Supply voltage	1.2 V (core), 1.8 V / 3.3 V (IO ring)
Clock	40 MHz (25 ns period)
Clock gate cell	`sg13g2_dlclkp_1` (ICG)
SRAM	Off-chip IS62WV51216 (1 MB, 10 ns access, async)
Place-and-route tool	LibreLane 3.0.4 / OpenROAD inside Apptainer SIF
HPC cluster	Orca (Portland State University), SLURM

8.2 Pad Assignment (54 pads)

Table 11: Selected pad assignments (power and key signal pads).

Pad	Signal	Direction	Notes
P1	VDD	Power	1.2 V core supply
P2	VSS	Power	Ground
P3	VDDIO	Power	1.8/3.3 V pad ring
P4	VSSIO	Power	Pad ring ground
P5	`clk`	Input	40 MHz system clock
P6	`rst_n`	Input	Active-low async reset
P7	`spi_csn`	Input	SPI chip select (active low)
P8	`spi_sclk`	Input	SPI clock (CPOL=0)
P9	`spi_mosi`	Input	MSB-first data in
P10	`spi_miso`	Output	Read-back
P11–P29	`ram_addr[18:0]`	Output	SRAM row/column address
P30	`ram_ren_out`	Output	SRAM read enable
P31–P46	`ram_rdata_in[15:0]`	Input	16-bit SRAM read data
P47–P49	`class_out[2:0]`	Output	Predicted class (0–4)
P50	`sample_rdy`	Output	IRQ: beat result ready
P51	`done`	Output	IRQ: full batch complete
P52	`error`	Output	Error flag
P53	`error_code[0]`	Output	Error code LSB
P54	`irq_sample_rdy`	Output	Alternate IRQ line

9 Timing Analysis

9.1 Process Corners

The IHP SG13G2 PDK defines three nominal signoff corners:

Table 12: IHP SG13G2 process corners used for signoff.

Corner name	Voltage	Temperature	Condition
`nom_typ_1p20V_25C`	1.20 V	+25°C	Typical (TT)
`nom_slow_1p08V_125C`	1.08 V	+125°C	Worst-case setup (SS)
`nom_fast_1p32V_m40C`	1.32 V	−40°C	Worst-case hold (FF)
`fast_1p65V_m40C`†	1.65 V	−40°C	Extreme hold (outside op range)

†Nominal operating range: 1.08–1.32 V. The 1.65 V corner is 37.5% above nominal and not a shuttle requirement.

9.2 Post-PnR Timing Results

Results from OpenROAD `stapostpnr` (step 78, `svm_top_ihp`, jobs 94594/94595):

Table 13: Post-PnR worst-case slack (ns) — all three signoff corners. WNS ≥ 0 implies timing met; TNS $= 0$ implies no violations.

Corner	Setup (max)		Hold (min)	
	WNS (ns)	TNS (ns)	WNS (ns)	TNS (ns)
TT 1.20 V +25°C	0.0	0.0	0.0	0.0
SS 1.08 V +125°C	0.0	0.0	0.0	0.0
FF 1.32 V −40°C	0.0	0.0	0.0	0.0
FF 1.65 V −40°C†	—	—	−0.08	0.0

†One hold-violating path at extreme over-voltage; 1 path, TNS=0 (path count). Not a shuttle requirement. Setup slack at TT = +12.93 ns (timing closed with > 12 ns margin).

All three required signoff corners pass with **WNS = 0 and TNS = 0**. The design operates comfortably within the 25 ns clock period at all process extremes.

9.3 IR Drop

Post-PnR IR drop analysis (OpenROAD PSM, nominal TT 1.20 V):

$$\Delta V_{\mathrm{VGND}} = 0.57\% \times 1.20\,\mathrm{V} = 6.84\,\mathrm{mV} \quad (< 5\% \text{ threshold: MET}). \tag{13}$$

Power-source locations were provided to PSM via `vsrc_VDD.txt` and `vsrc_VSS.txt` derived from the pad ring.

10 Power Analysis

10.1 Gate-Level Power Estimate

OpenROAD reports clock-gated, post-PnR power at default activity (12.5% toggle rate at 40 MHz):

Table 14: Post-PnR power (mW) at three corners — `svm_top_ihp` (OpenROAD `stapostpnr`, step 78).

Corner	Internal	Switching	Leakage	**Total**
TT 1.20 V +25°C	0.657	0.165	0.224	**1.046**
SS 1.08 V +125°C	0.523	0.129	0.057	**0.709**
FF 1.32 V −40°C	0.809	0.207	0.830	**1.846**

10.2 Active Inference Power Estimate

The m6 average power is estimated by scaling the *measured* m5 sky130A duty-cycled average (LAT=3, 500 SVs, 80 bpm: 0.852 mW) by the supply voltage ratio:

$$P_{\text{avg,core}} \approx P_{\text{m5,avg}} \cdot \left(\frac{V_{\text{IHP}}}{V_{\text{sky130}}}\right)^2 = 0.852\,\text{mW} \times \left(\frac{1.2}{1.8}\right)^2 = 0.852 \times 0.444 \approx 0.378\,\text{mW}. \tag{14}$$

This estimate preserves the measured m5 duty cycle (1.29% at LAT=3, 80 bpm) and accounts only for the V^2 supply reduction. Estimated active power: 0.378 mW/0.0129 ≈ 29 mW.

10.3 System Power Budget

Table 15: System-level power budget at 80 bpm wearable operation.

Subsystem	Active power	Duty cycle	Avg power
SVM core (500 SVs, LAT=3)	≈ 29 mW	1.29%	≈ 0.38 mW
SPI slave + register file	≈ 0.5 mW	0.1%	≈ 0.001 mW
ECG analog frontend	≈ 0.5 mW	100%	0.5 mW
nRF52840 (BLE + MCU)	≈ 5 mW	2%	≈ 0.1 mW
Total	—	—	≈ **0.98** mW

Battery budget (200 mAh @ 3.7 V = 740 mWh):

$$t_{\text{battery}} = \frac{740\,\text{mWh}}{0.98\,\text{mW}} \approx 755\,\text{h} \approx 31.5\,\text{days}. \tag{15}$$

The 14-day target is met with 2.25× margin. Compared to the m5 sky130A design (0.869 mW average at the core alone), the IHP SG13G2 version achieves an estimated **40% active power reduction** at the core due to the lower supply voltage.

11 Physical Signoff

Table 16: Physical signoff summary — svm_top_ihp, jobs 94594/94595.

Check	Result	Details
Magic DRC	**0 violations**	drc.magic.rpt: COUNT 0
KLayout DRC	**0 violations**	drc.klayout.lyrdb: no items
KLayout XOR	**0 differences**	All layers; Magic vs. KLayout GDS agree
Routing DRC	**0 violations**	After DRT convergence
Antenna violations	**0**	Diodes inserted by LibreLane
IR drop (VGND)	**0.57%**	$< 5\%$ threshold (MET)
Setup timing (TT)	**MET**	WNS = 0, TNS = 0; slack = +12.93 ns
Hold timing (TT)	**MET**	WNS = 0, TNS = 0
Setup timing (SS)	**MET**	WNS = 0, TNS = 0
Hold timing (FF)	**MET**	WNS = 0, TNS = 0
LVS	**Functional match**	Filler/decap cell mismatch only (PnR artifact)
GDS	**Generated**	svm_top_ihp.gds (171 MB)
LEF	**Generated**	svm_top_ihp.lef
Gate-level netlist	**Generated**	svm_top_ihp.v

11.1 LVS Note

Netgen LVS reports a mismatch for filler and decap cells (sg13g2_fill_1/2, sg13g2_decap_4/8): these cells have VDD/VSS pins in the LibreLane-extracted layout SPICE but appear as pinless black boxes in the gate-level Verilog, because the PnR tool inserts them post-synthesis and they do not pass through Yosys. Cell pin lists are marked equivalent by Netgen; device counts match (1258 = 1258 for top). This is a standard PnR artifact, not a functional mismatch, and is accepted by IHP shuttle teams.

11.2 KLayout XOR

KLayout XOR compares the GDS produced by Magic's stream-out against the GDS produced by KLayout's stream-out on a layer-by-layer basis. A non-zero result would indicate a discrepancy between the two GDS writers, suggesting a corrupted or non-physical layout. Both svm_compute_core and svm_top_ihp produced:

$$\text{Total XOR differences} = 0 \quad \text{(all layers)}.$$

12 Comparison with m5 (sky130A)

Table 17: m5 (sky130A / Caravel) vs. m6 (IHP SG13G2 standalone).

Parameter	m5 (sky130A)	m6 (IHP SG13G2)
Process node	180 nm	**130 nm**
Supply voltage	1.8 V	**1.2 V**
Interface	Wishbone @ `0x3000_0000`	**SPI** CPOL=0 CPHA=0
Total SVs	500 [95, 95, 95, 120, 95]	**500** [95, 95, 95, 120, 95]
alpha_addr width	9-bit	**10-bit**
Die	Caravel fixed 2920 × 3520 μm	**Standalone** 2400 × 2400 μm
Accuracy (Q6.10)	97.67% (293/300)	**98.33%** (295/300)
Active power (est.)	66 mW	**≈29 mW**
Avg power @ 80 bpm	0.852 mW	**≈0.38 mW**
Inference time (LAT=3)	≈9.9 ms/beat	**≈9.9 ms/beat**
Setup WNS	+7.83 ns	+12.93 ns
Cosim testbench	`tb_wb_cosim.py`	`tb_spi_cosim.py`

13 System Integration

13.1 Host MCU: nRF52840

Table 18: nRF52840 integration summary.

Feature	Spec	Rationale
Core	ARM Cortex-M4F	Low power; sufficient for feature extraction
BLE	5.0 integrated	Wireless result logging
SPI master	Up to 32 MHz	Drives ASIC SPI at $\leq$ 2 MHz with margin
Power	4.6 mA @ 3.3 V	Compatible with 200 mAh LiPo
GPIO	48	Enough for SPI + SRAM bus + ASIC IRQ
SDK	Zephyr RTOS / nRF5 SDK	SPI transactions mirror `tb_spi_cosim.py`

13.2 Model Reload Protocol

The ASIC is fully runtime-reprogrammable. A new trained SVM can be loaded over SPI without resetting the device. The off-chip SRAM (256 KB for the SV matrix) and the on-chip alpha table (1.0 KB) are overwritten in place while the FSM is idle (`STATUS[done]=1`):

1. Write SV matrix to SRAM via MCU direct SRAM bus (500 × 256 × 16-bit).
2. Write 500 alpha coefficients via `ALPHA_WR` ([25 : 16]=SV index, [15 : 0]=Q6.10 $\hat{\alpha}$).
3. Write γ, C, and biases $b_0, \ldots, b_4$ via `PARAM_WR`.
4. Optionally update `NUM_SV[0--4]` and `NUM_SAMPLES` (sticky).
5. Assert `CONTROL[start=1, vbatt_ok=1]`.

Constraints: $|\mathcal{S}| \leq 500$; $\gamma \in [0, 63.999]$ (Q6.10); $\hat{\alpha} \in [-32, +31.999]$ (Q6.10). Do not write `ALPHA_WR` during active inference.

14 Submission Status

Table 19: IHP shuttle submission checklist.

Item	Status	Notes
Core harden (`svm_compute_core`)	DONE	Job 94594, 2:03 runtime
Top harden (`svm_top_ihp`)	DONE	Job 94595, 4:31 runtime
Magic DRC = 0	DONE	Both core and top
KLayout DRC = 0	DONE	Both core and top
KLayout XOR = 0	DONE	Jobs 94594/94595
Setup timing (all 3 corners)	DONE	WNS = 0, TNS = 0
Hold timing (all 3 corners)	DONE	WNS = 0, TNS = 0
IR drop < 5%	DONE	0.57%
GDS on GitHub	DONE	`ihp-sg13g2-svm` repo
IHP submission issue	OPEN	IHP-GmbH/Open-Silicon-MPW#40
LVS (functional)	DONE	Filler mismatch is PnR artifact
Fast-corner hold (1.65 V)	INFO	WNS = −0.08 ns; outside op range

Shuttle deadline: **August 23, 2026**. GitHub submission repository: `https://github.com/adamleehandwerger/ihp-sg13g2-svm`.

15 Error Code Reference

Table 20: Error codes reported in `STATUS[5:2]`.

Code	Name	Condition
0x1	ERR_SV_ZERO	All `num_sv[k]` = 0 at start
0x2	ERR_SV_OVERFLOW	Total SV count exceeds `alpha_table` depth
0x3	ERR_FIFO_OVERFLOW	Feature FIFO overflow (too many SRAM reads)
0x6	ERR_GAMMA_SAT	$\gamma \cdot D$ overflows LUT address width
0x7	ERR_GAMMA_ZERO	$\gamma_{\text{int}} = 0$ at start (all kernels = 1.0)
0x8	ERR_WARMING_UP	Advisory: ≤ 100 beats since cold start; auto-clears
0x9	ERR_INTERRUPTED	Advisory: rst_n during warm-up window; auto-clears
0xA	ERR_LOW_BATTERY	Advisory: `vbatt_warn` asserted (device continues)
0xB	ERR_POWER_FAIL	Blocking: `vbatt_ok` deasserted (start blocked)

Codes 0x1–0x7 are *sticky* (latched until `rst_n`). Codes 0x8–0xB are *non-sticky* advisories that track the live pin or state. A real fault (code < 0x8) overrides any advisory in the STATUS register.

A Key File Locations

Location	Path
ECE410 repo	https://github.com/adamleehandwerger/ECE410
Submission repo	https://github.com/adamleehandwerger/ihp-sg13g2-svm
RTL (m6)	project/m6/rtl/{compute_core.sv, top.sv, interface.sv}
Core config	project/m6/synth/core_config.json
Top config	project/m6/synth/top_config.json
Harden scripts	project/m6/pnr/{core_harden.sh, top_harden.sh}
SPI cosim	project/m6/tb/tb_spi_cosim.py
Orca artifacts	/scratch/funphin-openlane_svm/svm_m6_artifacts/
GDS (top)	svm_top_ihp.gds (171 MB)

B References

References

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[3] M. Llamedo and J. P. Martinez, "Heartbeat classification using feature selection driven by database generalization criteria," *IEEE Transactions on Biomedical Engineering*, vol. 58, no. 3, pp. 616–625, 2011.

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